Free Energies of Hydration from a Generalized Born Model and an All-Atom Force Field

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The generalized Born/surface area (GB/SA) model of Still and co-workers was originally developed using partial atomic charges for organic molecules and ions from the OPLS united-atom force field. An efficient implementation of the GB/SA approach with the OPLS-AA (all-atom) force field is described here. Migration to the OPLS-AA model allows much broader application, and it also yields improved accuracy in predicting free energies of hydration. For 75 diverse, neutral organic molecules, the mean unsigned error is 0.6 kcal/mol with the OPLS-AA GB/SA model. Furthermore, effects of hydration on conformational equilibria are shown to be well represented, and results for free energies of hydration of a wide variety of ions are also in close accord with experimental data. As an even more general alternative, the use of partial charges from the CM1A procedure of Cramer, Truhlar, and co-workers has been tested on more than 400 organic molecules and ions. OPLS-AA force field parameters are also reported for primary alkyl halides, halobenzenes, and numerous ions.

Background

Inclusion of the effects of hydration is necessary for accurate characterization of the structures and energetics of molecular systems in aqueous solution. This is approached computationally by either explicit representation of the water molecules or through continuum models of the solvent.[1] The continuum approaches have enormous speed advantages and can allow more extensive sampling of solute structures in many contexts including conformational search, Monte Carlo (MC) statistical mechanics, and molecular dynamics (MD) simulations. For studies of organic and biomolecular systems, the generalized Born/surface area (GB/SA) model has emerged as the most popular continuum approach.[2,3] It is computationally relatively efficient and it provides excellent correlations with results obtained from solving the Poisson−Boltzmann equation, a more rigorous treatment of the electrostatic contributions to solvation.[4] Recent successes include folding of polypeptides to observed native structures by using MD[5−7] or MC[8,9] with GB/SA.

The GB/SA method was originally introduced by Still et al.[2] The free energy of hydration, ΔG_{hyd}, for a molecular system with atoms k and associated charges, q_k, and solvent-accessible surface areas, SA_k, is approximated by eqs 1−3. σ_k is an empirical solvation parameter for the cavity formation plus van der Waals (nonpolar) term that was fixed at 0.0072 kcal/(mol·Å^2) for all united atoms,[2] and the dielectric constant of water, ϵ, was set at 78.3. The double sum is over all atoms k and j, which are separated by a distance r_{kj}, and $\alpha_{kj} = (\alpha_k\alpha_j)^{1/2}$, where α_k is the Born radius, approximately the average distance of atom k from the dielectric boundary. van der Waals radii, R_k^{vdW}, also need to be specified and are used in the computation of

$$\Delta G_{hyd} = G_{cav} + G_{vdW} + G_{pol} \tag{1}$$

$$G_{cav} + G_{vdW} = G_{np} = \sum_k \sigma_k SA_k \tag{2}$$

$$G_{pol} = -\frac{1}{2}\left(1 - \frac{1}{\epsilon}\right)\sum_k\sum_j q_k q_j/(r_{kj}^2 + \alpha_{kj}^2 \exp(-r_{kj}^2/F\alpha_{kj}^2))^{1/2} \tag{3}$$

the Born radii and the surface areas. In the original paper, the q_k and R_k^{vdW} were taken from the OPLS united-atom (UA) force field with $R_k^{vdW} = (1/2)\sigma_k^{LJ}$, where σ_k^{LJ} is the Lennard-Jones σ for k, and $F = 4$. The predictions from eq 3 were shown to reproduce well the results of free-energy perturbation calculations for charging 20 small molecules and ions in explicit TIP4P water. Furthermore, the predictions from eq 1 for 20 neutral organic molecules correlated very well ($r^2 = 0.96$) with experimental ΔG_{hyd} values. In a subsequent study,[3] Still and co-workers simplified the computation of the Born radii through an analytical procedure, which features five adjustable parameters, P_1−P_5, as detailed below; some variation in σ_k for different atom types was also allowed. The OPLS-UA model continued to provide the q_k and R_k^{vdW} values, while validation featured comparison of GB/SA polarization energies with Poisson−Boltzmann results and computation of free energies of hydration for 35 neutral molecules.

Since that time, application of GB/SA methodology has largely focused on macromolecular systems.[4−14] Papers by Friesner, Levy, and co-workers are an exception and feature their surface generalized Born (SGB) modification for computing the Born radii via a surface integral.[15−18] In a recent report,[17] expansion of the nonpolar term (eq 2) was considered; the best

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results were obtained with eq 4, where γ and α are adjustable

$$G_{\text{np}} = \sum_k [\gamma(t_k)\text{SA}_k + \alpha(t_k)] \quad (4)$$

parameters dependent on atom types t_k, which come from the OPLS-AA force field along with the atomic charges. The α parameters were introduced to better reflect attractive (vdW) elements of the nonpolar term. However, their use for macromolecules with buried atoms is problematic and requires further modification with a switching function.[17] In all, 17 atom types were identified for neutral molecules resulting in 34 optimizable parameters for eq 4. For 99 molecules in the training set, the average unsigned error was 0.35 kcal/mol, which rose to 0.59 kcal/mol for the 93 molecules in a test set. In their latest study, another alternative was proposed for the vdW term with use of a similar number of atom types and parameters.[18] The extensive efforts of Cramer, Truhlar, and co-workers should also be noted.[19,20] Their approach features an extensively modified and parametrized GB/SA framework in conjunction with computation of the atomic charges, typically from semiempirical quantum mechanics. The introduction of so many parameters moves these methods away from the simplicity of the original GB/SA procedure[2,3] and more toward fragment-based regression models.[21] In the present paper, we share our experiences in the implementation of GB/SA with the OPLS-AA force field[22,23] in a manner more consistent with the original model.

Computational Details

Our approach follows the implementation of the analytical model of Qiu et al.[3] Equations 1−3 apply and the Born radii, α_k, come via the Born equation from evaluation of the polarization energy for atom k embedded in the full solute according to eqs 5 and 6. The polarization energy for k is made

$$G_{\text{pol}}{}^{k} = -1/(2\alpha_k) = -1/(2\gamma_k) + \sum_j^{1,2} P_2 V_j/r_{kj}{}^4 + \sum_j^{1,3} P_3 V_j/r_{kj}{}^4 + \sum_j^{\geq 1,4} P_4 V_j C_{kj}/r_{kj}{}^4 \quad (5)$$

$$\gamma_k = R_k{}^{\text{vdW}} + \varphi + P_1 \quad (6)$$

less favorable by the screening from the other atoms in the solute, and the summations in eq 5 reflect the contributions from bonded neighbors, 1,3-neighbors, and more distant atoms. V_j is the volume of atom j and C_{kj} is the "close contact function" (CCF), as defined in ref 3. The P_i are adjustable parameters; the remaining one, P_5, is used to define the cutoff distance for the CCF such that $C_{kj} = 1$, if $r_{kj}{}^2/(R_k{}^{\text{vdW}} + R_j{}^{\text{vdW}})^2 > 1/P_5$. Finally, φ is the dielectric offset, which was fixed at −0.09 Å.[2,3] Reoptimization of φ and the P_i was pursued with the all-atom model, but very modest error reductions were obtained, so their original values[3] have been retained here.

There are additional details that need to be specified. (1) As before,[2,3] $R_k{}^{\text{vdW}} = (1/2)\sigma_k{}^{\text{LJ}}$ and $R_k{}^{\text{vdW}}$ for a hydrogen bonded to a heteroatom is set at 1.15 Å. In addition, we set $R_k{}^{\text{vdW}} =$ 2.0 Å for covalent fluorine as the only other exception. Halogens were not considered in the original papers,[2,3] and we have not previously reported the OPLS-AA parameters for primary alkyl halides or halobenzenes. This is now done for the nonbonded parameters in Tables 1 and 2; validation has included the usual excellent reproduction of experimental liquid densities and heats of vaporization.[22,23] (2) All hydrogens are included in the calculations of the atomic volumes V_j, and all hydrogens are ignored in the computations of the solvent accessible surface areas for eq 2. The atomic radii for the SA calculations are taken as $\sigma_k{}^{\text{LJ}}/2$, and the radius of the water probe is 1.4 Å.[2,3] (3) The σ_k parameters for eq 2 were adopted from ref 3 except we leave sulfur with the generic 0.007 instead of 0.010 kcal/(mol·Å^2), which has negligible effect on the present results. Values then had to be added for the halogens. This required some optimization, and the final choices are included in Table 3. The range for the halogens reflects their range in the periodic table, and the range and trend for the ϵ^{LJ} values in Tables 1 and 2. (4) For eqs 3 and 5, a united-atom simplification is again adopted for hydrogens on saturated carbon. Thus, the charges on these hydrogens are added to the charge on the adjacent carbon, and the hydrogens are then ignored. This significantly reduces the number of nonbonded pairs that need consideration in eq 5. All other hydrogens including those on unsaturated carbons are treated in the same manner as heavy atoms. (5) In evaluating the last two summations in eq 5, k,j-pairs are excluded if k and

TABLE 1: OPLS-AA Nonbonded Parameters and GB/SA van der Waals Radii for Alkyl Halides, RCH_2X[a]

atom[b]	q, e	σ, Å	ϵ, kcal/mol	R^{vdW},[c] Å
F	−0.220	2.94	0.061	2.000
Cl	−0.200	3.40	0.300	1.700
Br	−0.200	3.47	0.470	1.735
I	−0.130	3.75	0.600	1.875
CT(F)	0.020	3.50	0.066	1.750
CT(Cl)	−0.006	3.50	0.066	1.750
CT(Br)	−0.006	3.50	0.066	1.750
CT(I)	−0.070	3.50	0.066	1.750
HC(F)	0.100	2.50	0.030	1.250
HC(Cl)	0.103	2.50	0.030	1.250
HC(Br)	0.103	2.50	0.030	1.250
HC(I)	0.100	2.50	0.030	1.250

[a] R is neutral and uses OPLS-AA parameters for alkanes. [b] CT is the AMBER/OPLS atom type for a saturated carbon. HC(X) is a hydrogen on the carbon α to X. [c] $R^{\text{vdw}} = \sigma/2$ except for F and for H on a heteroatom (1.15 Å).

TABLE 2: OPLS-AA Nonbonded Parameters and GB/SA van der Waals Radii for Halobenzenes

atom[a]	q, e	σ, Å	ϵ, kcal/mol	R^{vdW}, Å
F[b]	−0.220	2.85	0.061	2.000
Cl	−0.180	3.40	0.300	1.700
Br	−0.200	3.47	0.470	1.735
I	−0.100	3.75	0.600	1.875
CA(F)	0.220	3.55	0.070	1.775
CA(Cl)	0.180	3.55	0.070	1.775
CA(Br)	0.200	3.55	0.070	1.775
CA(I)	0.100	3.55	0.070	1.775
CA	−0.115	3.55	0.070	1.775
HA	0.115	2.42	0.030	1.250

[a] CA is the AMBER/OPLS atom type for a carbon atom in a six-membered aromatic ring and HA is an attached hydrogen. [b] For difluorobenzenes, $q_F = -0.20$; for hexafluorobenzene, $q_F = -0.13$. Parameters for other polyhalobenzenes have not yet been optimized.

TABLE 3: GB/SA σ_k Atomic Solvation Parameters (kcal/(mol·Å^2))

atom[a]	σ_k
H, N, O	0.000
CT, F	0.010
Cl, other	0.007
Br	0.003
I	−0.012

[a] CT is the atom type for a saturated carbon.

TABLE 4: Experimental and GB/SA Free Energies of Hydration (kcal/mol)

molecule	exptl[a]	OPLS-AA	1.07·CM1A	Qiu[3]	molecule	exptl[a]	OPLS-AA	1.07·CM1A	Qiu[3]
propane	1.96	1.82	1.82	1.8	anisole	−2.46	−2.40	−3.70	
butane	2.07	2.10	2.10	2.1	morpholine	−7.18	−6.36	−4.75	−7.7
hexane	2.48	2.65	2.65	2.5	aniline	−5.49	−4.85	−7.09	−4.1
octane	2.88	3.20	3.20	2.9	3-methoxyaniline	−7.30	−6.11	−8.39	
2,4-dimethylpentane	2.84	2.80	2.79	2.8	dimethylamine	−4.30	−4.30	−3.79	−2.3
cyclohexane	1.23	2.28	2.28	1.8	aziridine	−5.41	−6.15	−3.20	
but-1-yne	−0.16	0.40	−1.66		piperazine	−7.37	−7.89	−6.42	
isoprene	0.68	0.66	0.28		trimethylamine	−3.22	−3.07	−1.44	
1,2-dichloroethane	−1.79	−2.01	−0.58		*N*-methylpiperidine	−3.78	−1.80	−0.60	
1-bromopentane	−0.10	0.18	0.97		pyridine	−4.69	−4.46	−5.20	−4.2
1-iodopropane	−0.53	−0.58	−1.39		2-methylpyrazine	−5.51	−6.18	−5.26	−5.0
benzene	−0.86	−1.10	−2.32	−1.0	acetone	−3.81	−4.34	−2.68	−2.7
toluene	−0.89	−0.97	−1.68	0.0	butanone	−3.71	−3.70	−1.89	−2.0
o-xylene	−0.90	−0.96	−1.07	1.0	acetophenone	−4.58	−4.85	−5.23	
naphthalene	−2.40	−1.81	−4.20	−1.5	benzaldehyde	−4.03	−4.95	−6.63	−5.5
PhF	−0.81	−0.48	−1.76		*p*-hydroxybenzaldehyde	−10.47	−10.93	−11.50	
$PhCF_3$	−0.25	−0.52	−1.41		acetic acid	−6.70	−7.49	−7.32	−6.4
PhCl	−1.12	−0.90	−2.23		methyl actetate	−3.14	−2.74	−4.01	−2.1
PhBr	−1.45	−1.45	−2.78		ethyl benzoate	−3.64	−2.99	−6.03	
PhI	−1.75	−1.85	−3.96		acetonitrile	−3.85	−8.53	−4.73	
methanol	−5.10	−6.31	−4.19	−7.3	acetonitrile UA	−3.85	−4.89		−4.7
ethanol	−5.01	−5.67	−3.68	−5.6	benzonitrile	−4.22	−3.99	−5.34	
2-propanol	−4.75	−5.07	−3.14	−4.3	*p*-cyanophenol	−10.18	−9.93	−10.26	
1-butanol	−4.72	−5.02	−3.13	−5.0	acetamide	−9.71[b]	−11.66	−10.50	−11.2
tert-butanol	−4.48	−4.50	−2.67		*N*-methylacetamide	−10.08[b]	−8.06	−10.39	−8.1
1-hexanol	−4.41	−4.45	−2.57	−4.6	dimethylformamide	−7.80	−6.99	−10.06	
cyclohexanol	−5.47	−4.47	−2.64		dimethylacetamide	−8.55[b]	−7.24	−8.01	−5.7
phenol	−6.62	−6.75	−7.04	−6.4	benzamide	−11.01	−12.86	−13.11	
o-chlorophenol	−4.56	−5.07	−5.72		nitroethane	−3.71	−7.29	−9.95	
o-iodophenol	−6.21	−6.25	−7.33		nitrobenzene	−4.12	−5.43	−10.91	
p-fluorophenol	−6.20	−6.33	−6.43		*m*-nitroaniline	−8.86	−9.39	−15.72	
p-chlorophenol	−7.04	−6.75	−7.00		*p*-nitrophenol	−10.65	−11.46	−15.89	
p-bromophenol	−7.13	−7.34	−7.53		ethyl mercaptan	−1.15	−1.00	0.20	−2.1
1-naphthol	−7.68	−7.05	−8.58		ethyl methyl sulfide	−1.50	−2.90	0.99	−2.0
2-phenylethanol	−6.80	−6.97	−6.20		diethyl disulfide	−1.64	−2.11	1.15	
2-ethoxyethanol	−6.70	−6.89	−5.22		dimethyl sulfoxide	−8.61	−8.20	−14.91	
dimethyl ether	−1.91	−2.36	−1.43	−3.8	thiophenol	−2.55	−1.68	−3.43	−2.8
1,4-dioxane	−5.05	−4.47	−2.92		thioanisole	−2.73	−2.39	−2.34	−2.0
r^2 vs expt		0.93	0.77	0.89					
mean unsigned error		0.61	1.51	0.88					
r^2 (no nitro, DMSO)		0.93	0.85						
mue (no nitro, DMSO)		0.56	1.16						

[a] Reference 25. [b] Reference 26.

j are directly bonded or, for the last term, if they have a common neighbor, i.e., they are 1,3-related.

In summary, the OPLS-UA based GB/SA approach of Qiu et al.[3] has been adopted with minor change for use with the OPLS-AA force field. A united-atom simplification continues to be used for hydrogens on saturated carbon for eqs 3 and 5, though all hydrogens are included in the calculation of the atomic volumes, V_j. Atom-typing is not required in this implementation; a count of bonded neighbors is sufficient to decide if a carbon atom is saturated. Our principal extension has been to add the treatment of halogens, as specified in Tables 1−3, and to validate the model more extensively for organic molecules and ions. For application to macromolecular systems, modifications to F in eq 3 and to the calculation of the Born radii are common to adjust for deviations from the Coulomb field approximation that is inherent in eq 5. Advances in this area have led to impressive comparisons between results of Poisson−Boltzmann and GB/SA calculations for proteins and nucleic acids.[4,10−16]

Results and Discussion

Free Energies of Hydration for Neutral Molecules. Free energies of hydration have been computed by performing a full energy minimization for a molecule including the GB/SA contribution, ΔG_{hyd} (eq 1), during the optimizations with use of the BOSS program.[24] Inclusion of the GB/SA term has little effect on the molecular geometries and internal energies. For example, the internal energies of *N*-methylacetamide and 1-hexanol increase by 0.06 and 0.10 kcal/mol, respectively, upon GB/SA hydration. As usual, the internal energy change has not been included in the reported results, i.e., the reported ΔG_{hyd} corresponds to the value computed via eq 1 for the relaxed solution-phase geometry. Results are provided in Table 4 for the 35 molecules from the paper by Qiu et al. plus an additional 40 molecules that have been selected for their variety and potential difficulty. The latter include examples of halogenated molecules, an alkyne, a diene, a tertiary alcohol, tertiary amines, many difunctional molecules, nitro compounds, a disulfide, and a sulfoxide, which were not included in the original study. The computed and experimental values are for a temperature of 25 °C and 1 M standard states. The results with the OPLS-AA model are also compared with the experimental data in Figure 1.

For the 75 molecules, the mean unsigned error (mue) for the GB/SA results with the OPLS-AA force field is 0.61 kcal/mol and the correlation coefficient r^2 is 0.93. For the 35 molecules

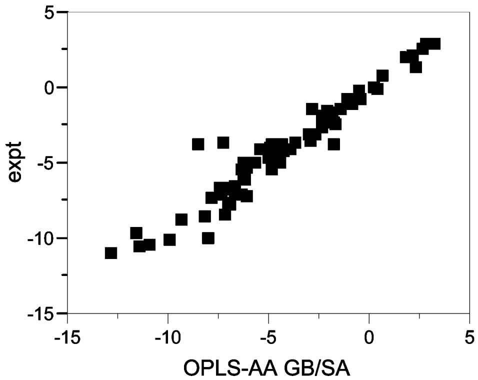


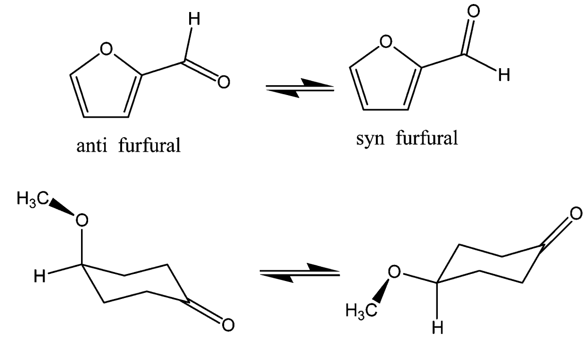
Figure 1. Computed vs experimental free energies of hydration (kcal/mol) for the 75 molecules in Table 4.

from ref 3, the mue and r^2 from Qiu et al. are 0.88 kcal/mol and 0.89, while the corresponding values are 0.67 kcal/mol and 0.91 for this subset with OPLS-AA. Thus, progression from the UA-based procedure to the present AA one does lead to significant improvement in the predicted free energies of hydration. The overall results are also impressive given that the uncertainties in the experimental data are ca. $\pm$0.3 kcal/mol,[25,26] and the mue for ΔG_{hyd} is 0.69 kcal/mol from free-energy perturbation (FEP) calculations on 27 molecules with use of the OPLS-AA force field and explicit TIP4P water.[27] We attribute the remaining differences primarily to specific issues with the OPLS-AA charge distributions and model. There is no compelling reason to seek improvement in the GB/SA model.

In particular, there are only two striking outliers in the OPLS-AA results, acetonitrile and nitroethane. Amides are handled reasonably well, though improvements for secondary amides can be envisioned.[27] In the Qiu et al. model,[3] the error for acetonitrile is 0.9 kcal/mol, which rises to 3.8 kcal/mol with the AA approach. The change comes from the difference in UA and AA charge distributions. In the OPLS-UA model, the nitrile C and N charges are 0.28 and $-$0.43,[28] and they are 0.46 and $-$0.56 in the OPLS-AA model.[23] The corresponding dipole moments for acetonitrile are 3.44 and 4.13 D. If one reverts to the UA charge distribution and applies the AA GB/SA model, the computed ΔG_{hyd} retreats to $-$4.89 kcal/mol, which is very close to the result of Qiu et al. (Table 4). We have confirmed our reported OPLS-AA results for liquid acetonitrile and propionitrile; the average errors for density and ΔH_{vap} are 2%.[23] The ΔG_{hyd} for acetonitrile was also recently recomputed via FEP calculations and is -4.1 ± 0.3 kcal/mol,[27] which is close to the experimental value of $-$3.9 kcal/mol. Though the OPLS-AA model for acetonitrile appears fine, it is likely that alternative charge sets can be devised without significant degradation in the explicit solvent results and that yield a lower dipole moment and improved GB/SA results. For GB/SA applications, one could simply use the OPLS-UA charges for alkyl nitriles; this change would reduce the mue error for the 75 molecules from 0.61 to 0.56 kcal/mol. It should also be noted that aryl nitriles are not problematic; the errors for benzonitrile and *p*-cyanophenol in Table 4 are only 0.2$-$0.3 kcal/mol. The OPLS-UA and OPLS-AA models are identical in this case with charges of 0.395 and $-$0.430 for nitrile C and N.[22b]

Treatment of nitro groups is generally problematic for classical molecular modeling.[17,20,27,29−31] Nitroalkanes typically emerge as too polar and hydrophilic. This is an extreme case in which the proximity of heteroatoms renders the simple model for the electrostatics of a partial charge on each atom inadequate. Nevertheless, the OPLS-AA force field manages to yield good results for density and ΔH_{vap} of pure nitroalkane liquids,[23] and the FEP-computed ΔG_{hyd} for nitroethane is -3.0 ± 0.5 kcal/mol.[27] However, the OPLS-AA charge distribution coupled with the GB/SA model leads to overly favorable hydration of nitroethane by 3.6 kcal/mol. The problem is ameliorated for nitroaromatics; the OPLS-AA approach for substituted benzenes is to merge the charges for benzene with those for the substituent. This increases the charge on the nitrogen by 0.11 to 0.65, but it partially neutralizes the charge on the attached hydrocarbon by going from 0.20 to 0.09. The net result is a lowering of the computed dipole moment from 3.75 D for nitroethane to 3.31 D for nitrobenzene; the experimental values are in the opposite order, 3.23 and 4.22 D. Thus, it appears that there are inconsistencies in the OPLS-AA charge distributions for nitro compounds. It is suspected that a reparametrization could bring both the explicit solvent and GB/SA results in-line with experimental data. Until this is resolved, for GB/SA applications, one could just replace the OPLS-AA charges for N and O of 0.54 and $-$0.37 with 0.30 and $-$0.25 for nitroalkanes. The result is a GB/SA ΔG_{hyd} for nitroethane of $-$3.63 kcal/mol.

SCHEME 1



In the alternative treatment with the OPLS-AA force field of Gallicchio et al.,[17] the best result is a mue of 0.35 kcal/mol for the training set of 99 molecules and a mue of 0.59 kcal/mol for the test set of 93 molecules.[32] This involves use of eq 4 and the associated introduction of 34 optimized parameters. If a constant γ is used with 17 optimized $\alpha(t_k)$ values, the corresponding mue results are 0.59 and 0.66 kcal/mol.[17] The former difference in mue values indicates over-fitting. However, the largest concern is for the range of the fitted γ and α values; γ ranges from $-$0.099 to 0.145 kcal/(mol·Å^2) and α ranges from $-$18.8 to 0.6 kcal/mol for the components of neutral molecules in Table 8 of their paper. Though some physical justification was provided for eq 4,[17] the range of the parameters negates this. Not surprisingly, some of the extreme values for γ and α are for nitrile and nitro functionality. Other large γ values are given for ether oxygens and nitrogens in amines and amides; this is likely a fitting artifact, which stems from the fact that these atoms usually have small solvent-accessible surface areas.

Conformational Equilibria. Effects of hydration on conformational equilibria were also studied for a few systems. The predictions are all qualitatively correct. For 1,2-difluoroethane and 1,2-dicholorethane, the gauche conformers are better hydrated than the anti conformers by 0.7 kcal/mol in both cases according to the OPLS-AA GB/SA calculations, while the experimental shift for 1,2-dicholorethane is 1.0 kcal/mol in going from the gas phase to acetonitrile solution.[33,34] And, for furfural (Scheme 1), the syn conformer is predicted to be better hydrated that the anti form by 1.9 kcal/mol according to the OPLS-AA GB/SA calculations, while the experimental differences are 1.55

and ca. 2 kcal/mol in going from the gas phase to acetone and DMSO.[34,35] The OPLS-AA and experimental[35] energy differences are 1.5 and 0.8 kcal/mol favoring the anti conformer in the gas phase, while the syn form is dominant in solvents more polar than hydrocarbons.[35] Finally, for 4-methoxycyclohexanone (Scheme 1), the axial form is computed to be lower in energy than the equatorial isomer in the gas phase by 1.0 kcal/mol, while the equatorial form is favored by 0.3 kcal/mol in water. This is also the pattern found by NMR experiments at 308 K, though the observed shift in going from cyclohexane to water, 0.54 kcal/mol, is not as large.[36] The computed dipole moments of 3.15 D for the equatorial form and 3.27 D for the axial conformer in the gas phase would lead to the opposite trend, so this is an interesting case of the dominance of higher moments that is captured in the GB/SA treatment.

Free Energies of Hydration for Ions. The present model was also extended to ions. OPLS-AA nonbonded parameters are collected in Table 5 for ammonium, imidazolium, guanidinium, *tert*-butyl, carboxylate, methoxide, methanethiolate, and atomic ions. The OPLS-AA parameters have been reported previously for most of the nitrogen-containing ions,[22a] carboxylate ions,[22a] and methoxide and methanethiolate,[37] and the parameters for the atomic cations have been taken from Åqvist.[38] Parameters for other ammonium and carboxylate ions can be readily assigned by incorporating standard OPLS-AA parameters for alkyl groups.[22a] The listed R^{vdW} values for the GB/SA calculations are mostly $(1/2)\sigma_k^{LJ}$, as above. The exceptions are the radii for hydrogens and carbons attached to ammonium nitrogens and for carboxylate oxygens, and the radii of atomic ions. These radii have been optimized to yield agreement with experimental free energies of hydration (Table 6). Given the optimization, there is no error for the atomic ions. Furthermore, for the polyatomic ions, the errors for the computed values are close to negligible given the uncertainties in experimental assignments for the free energies of hydration of single ions.

Some ion−ion potentials of mean force (pmfs) were also computed with the OPLS-AA plus GB/SA model. This was done by repetitive optimizations of all variables except the interionic distance as the ions are separated; for proper sampling of polyatomic ions at a specified temperature, MC or MD simulations should be run at each value of the interionic distance. The dissociation limits are correct since there is no error for the free energies of hydration of the individual ions. Examples are provided for tetramethylammonium chloride, guanidinium chloride, and sodium chloride in Figure 2. The former two ion pairs do not have minima in their pmfs, which is consistent with their known lack of association in water.[39,40] For sodium chloride, free energy calculations in explicit water find minima in the pmf for contact and solvent-separated ion pairs with well depths of 1−4 kcal/mol at separations of ca. 2.9 and 5.0 Å.[41,42] With the OPLS-AA GB/SA model, the well depth is 4.9 kcal/mol at a separation of 2.9 Å (Figure 2); the continuum model cannot yield a solvent-separated ion pair.

TABLE 5: OPLS-AA Nonbonded Parameters and GB/SA van der Waals Radii for Ions

atom[a]	ion	q, e	σ, Å	ϵ, kcal/mol	R^{vdW}, Å
N3	NH_4^+	−0.400	3.25	0.170	1.625
N3	RNH_3^+	−0.300	3.25	0.170	1.625
N3	$R_2NH_2^+$	−0.200	3.25	0.170	1.625
N3	R_3NH^+	−0.100	3.25	0.170	1.625
N3	R_4N^+	0.000	3.25	0.170	1.625
H	NH_4^+	0.350	0.00	0.000	1.350
H	RNH_3^+	0.330	0.00	0.000	1.350
H	$R_2NH_2^+$	0.310	0.00	0.000	1.350
H	R_3NH^+	0.290	0.00	0.000	1.350
CT	$CH_3NH_3^+$	0.130	3.50	0.066	2.100
CT	$RCH_2NH_3^+$	0.190	3.50	0.066	2.100
CT	$RCH_3NH_2^+$	0.110	3.50	0.066	2.100
CT	$R_2CH_3NH^+$	0.090	3.50	0.066	2.100
CT	$(CH_3)_4N^+$	0.070	3.50	0.066	2.100
HC	all ammonium	0.060	2.50	0.030	1.250
N3	$PhNH_3^+$	−0.300	3.25	0.170	1.625
H	$PhNH_3^+$	0.330	0.00	0.000	1.350
CA	$PhNH_3^+$	0.310	3.55	0.070	1.775
NA	imidazolium	−0.540	3.25	0.170	1.625
H	imidazolium	0.460	0.00	0.000	1.150
CR	imidazolium	0.385	3.55	0.070	1.775
CX	imidazolium	0.215	3.55	0.070	1.775
HA	imidazolium	0.115	2.42	0.030	1.210
N2	guanidinium	−0.800	3.25	0.170	1.625
H	guanidinium	0.460	0.00	0.000	1.150
C	guanidinium	0.640	3.55	0.076	1.775
C+	$(CH_3)_3C^+$	0.619	3.55	0.076	1.775
CT	$(CH_3)_3C^+$	−0.395	3.50	0.066	1.750
HC	$(CH_3)_3C^+$	0.174	2.50	0.030	1.250
O2	$RCOO^-$	−0.800	2.96	0.210	1.600
C	$RCOO^-$	0.700	3.75	0.105	1.825
CT	CH_3COO^-	−0.280	3.50	0.066	1.750
CT	RCH_2COO^-	−0.220	3.50	0.066	1.750
C	$PhCOO^-$	0.815	3.75	0.105	1.825
CA	$PhCOO^-$	−0.215	3.55	0.070	1.775
HC	all $RCOO^-$	0.060	2.50	0.030	1.250
O[b]	CH_3O^-	−0.980	3.15	0.250	1.575
CT	CH_3O^-	−0.200	4.20	0.300	2.100
HC	CH_3O^-	0.060	2.50	0.050	1.250
S	CH_3S^-	−0.900	4.25	0.500	2.125
CT	CH_3S^-	−0.400	4.20	0.300	2.100
HC	CH_3S^-	0.100	2.50	0.050	1.250
F	F^-	−1.000	2.733	0.720	1.540
Cl	Cl^-	−1.000	4.417	0.118	2.090
Br	Br^-	−1.000	4.624	0.090	2.255
I	I^-	−1.000	5.400	0.070	2.700
Li[c]	Li^+	1.000	2.126	0.01828	1.350
Na	Na^+	1.000	3.330	0.00277	1.680
K	K^+	1.000	4.935	0.00033	2.020
Rb	Rb^+	1.000	5.622	0.00017	2.160
Cs	Cs^+	1.000	6.716	0.00008	2.540
Mg	Mg^{2+}	2.000	1.644	0.87504	1.455
Ca	Ca^{2+}	2.000	2.412	0.44966	1.735
Sr	Sr^{2+}	2.000	3.103	0.11823	1.910
Ba	Ba^{2+}	2.000	3.817	0.04710	2.105

[a] N3 and N2 are ammonium and guanidinium nitrogens, O2 is a carboxylate oxygen. [b] q, σ, and ϵ for CH_3O^- and CH_3S^- are from ref 37. [c] q, σ, and ϵ for the atomic cations are from ref 38.

Results with CM1A Atomic Charges. OPLS-AA parameters have not been developed for all possible organic molecules and ions. Through atom-typing and the use of a hierarchy of estimation procedures, it is possible to assign automatically reasonable Lennard-Jones parameters and the parameters for bond stretching, angle, bending, and torsional energetics.[24] The principal problem is the assignment of the partial atomic charges for arbitrary molecules, which may contain, for example, a new heterocyclic ring. For this purpose, our preferred procedure is to generate the charges from a quantum mechanical wave function. Storer et al. reported the particularly attractive CM1A and CM1P approaches that provide high-quality partial charges from rapid, semiempirical AM1 and PM3 quantum calculations.[30] Their mapping procedures were optimized to yield accurate dipole moments for organic molecules. Though subsequent efforts led to the introduction of Charge Model 3 (CM3),[31] on the basis of FEP results for free energies of hydration, the CM1A charges emerge as preferred for use with the remaining parameters in the OPLS-AA force field.[27] Since the CM charges are optimized to reproduce gas-phase dipole moments, they are scaled upward for use in simulations in explicit water. For neutral molecules, the optimized scale factor

TABLE 6: Experimental and GB/SA Free Energies of Hydration (kcal/mol) for Ions

ion	exptl[a]	OPLS-AA	1.00·CM1A
NH_4^+	−86, −79	−81.8	−81.8
$CH_3NH_3^+$	−71, −73	−72.2	−73.1
$CH_3CH_2NH_3^+$	−68	−72.5	−71.4
$CH_3CH_2CH_2NH_3^+$	−67	−72.2	−70.7
$(CH_3)_2NH_2^+$	−64, −66	−65.1	−65.3
piperidinium	−60	−64.6	−62.3
$(CH_3)_3NH^+$	−57, −59	−57.4	−58.0
$(CH_3)_4N^+$	−52	−51.1	−51.4
$PhNH_3^+$	−66, −68	−71.8	−63.7
imidazolium	−64	−67.0	−69.5
guanidinium		−77.3	−77.1
$(CH_3)_3C^+$		−74.2	−68.2
CH_3COO^-	−80, −77	−79.6	−74.6
$CH_3CH_2COO^-$	−79	−78.6	−72.4
$PhCOO^-$	−76	−77.0	−67.2
CH_3O^-	−95	−94.0	−93.9
CH_3S^-	−76	−72.3	−86.9
F^-	−107	−106.7	−106.7
Cl^-	−77	−77.9	−77.9
Br^-	−72	−72.7	−72.7
I^-	−63	−63.6	−63.6
Li^+	−122	−122.5	−122.5
Na^+	−98	−97.8	−97.8
K^+	−81	−80.5	−80.5
Rb^+	−76	−75.8	−75.8
Cs^+	−68	−68.4	−68.4
Mg^{2+}	−456	−455.6	−455.6
Ca^{2+}	−381	−381.1	−381.1
Sr^{2+}	−346	−346.1	−346.1
Ba^{2+}	−315	−315.7	−315.7

[a] References 15, 18, and 38.

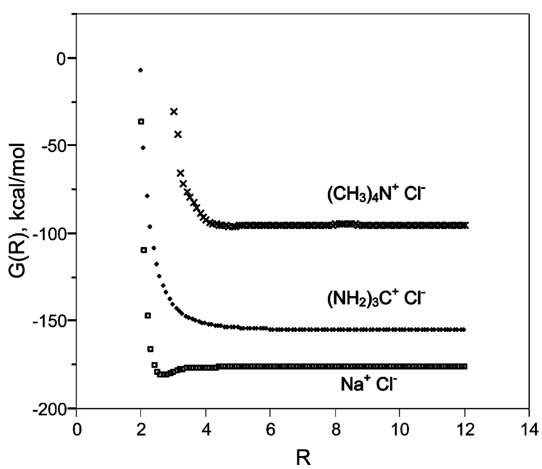


Figure 2. Ion−ion potentials of mean force for tetramethylammonium chloride (top), guanidinium chloride (middle), and sodium chloride (bottom). *R* is the interionic separation in Å; the N−Cl, C−Cl, and Na−Cl distances, respectively.

for the CM1A charges was determined to be 1.14 to minimize the error in computed free energies of hydration;[27] the resultant mue is 1.0 kcal/mol from the FEP results for 25 organic molecules in TIP4P water.[27] The atomic charges for ions are unscaled so that the net charge remains an integer.

So, for the GB/SA calculations, replacement of the OPLS-AA atomic charges with CM1A charges has been pursued. The scale factor of 1.14 for the CM1A charges from the FEP calculations in explicit water is not necessarily the optimal scale factor using GB/SA. This was examined for the 75 molecules in Table 4, and it was found that the optimal scale factor for neutral molecules with the present GB/SA model is 1.07. As listed in Table 4, the resultant mue is 1.51 kcal/mol and the r^2 is 0.77 for the 75 free energies of hydration; a quick scan shows that nitro compounds and DMSO are the big problems. If they are removed, the mue and r^2 improve to 1.16 kcal/mol and 0.85. Interestingly, nitriles are not a problem. For acetonitrile, the 1.07·CM1A dipole moment is 4.0 D and the charges on the methyl group and nitrile C and N are −0.22, 0.12, and −0.34, while they are 0.10, 0.46, and −0.56 with OPLS-AA. Again, it seems likely that the OPLS-AA charges could be reworked to yield improved results for the GB/SA calculations without degradation of results for explicit solvent simulations.

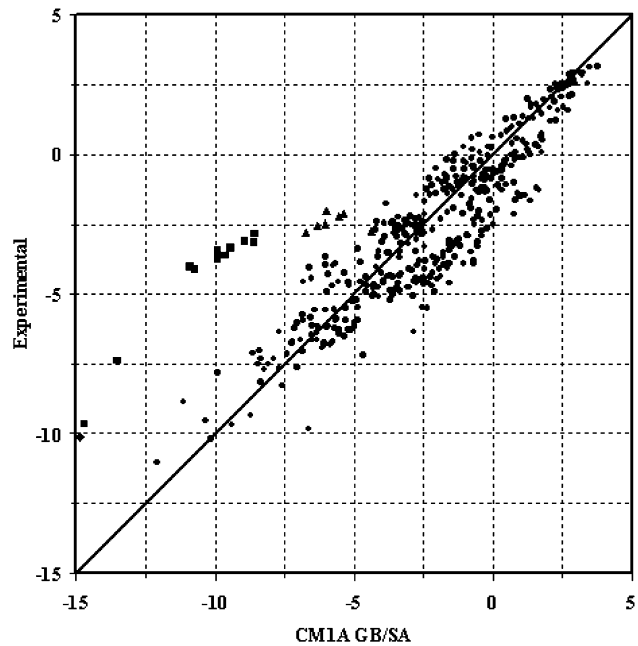


Figure 3. Computed vs experimental free energies of hydration (kcal/mol) for 420 neutral organic molecules from ref 25. Filled boxes are nitro compounds, triangles are formate esters, and the diamond at the far left is DMSO.

The CM1A GB/SA results for ion hydration are given in Table 6. For the atomic ions, there are no differences from the OPLS-AA results since the charges are fixed. However, there is also close accord between the OPLS-AA, 1.00·CM1A, and experimental results for the polyatomic ions. The principal differences are for the anilinium and benzoate ions, which are more charge delocalized with the CM1A approach, and for methanethiolate, which shows the opposite effect.

The GB/SA calculations with the 1.07·CM1A charges were then applied to the extensive dataset of Abraham et al.[25] In all, 420 polyatomic, neutral organic molecules were treated. Initial geometries were generated with a 3D structure generator from 2D representations; for flexible molecules, extended conformers, e.g., all trans, are produced. CM1A charges were then computed in a single-point calculation, followed by geometry optimization with the 1.14·CM1A/OPLS-AA force field, recomputation of the charges, and the single-point 1.07·CM1A GB/SA calculation. The results are illustrated in Figure 3; the only new problem that is revealed is for formate esters. None were included in Table 4, and they are not problematic with the OPLS-AA charges. For example, the experimental ΔG_{hyd} values for methyl formate and ethyl formate are −2.78 and −2.57 kcal/mol, and the OPLS-AA GB/SA results are −3.22 and −2.78 kcal/mol; however, the 1.07·CM1A charges yield −6.19 and −5.80 kcal/mol. If the nitro compounds, formates, and DMSO are excluded, the mue for the remaining 399 compounds is 1.01 kcal/mol.

Conclusion

The GB/SA procedure of Still and co-workers[2,3] has been extended in a straightforward manner for application with the OPLS-AA force field. In view of the similarity of many fixed-charge, all-atom force fields for bioorganic systems, it is likely that the present implementation can find broader use. The presented results for free energies of hydration of diverse organic molecules and polyatomic ions illustrate the good accuracy and generality of the OPLS-AA GB/SA approach. With a mean unsigned error of 0.6 kcal/mol for neutral molecules, the free energies of hydration from this computationally undemanding method are at least as accurate as those from full FEP calculations with the OPLS-AA force field in explicit water.[27] Introduction of more highly parametrized, more complex GB/SA procedures for organic molecules and ions is not justified. More global application has also been considered through use of CM1A charges. Though there is degradation in accuracy from the OPLS-AA level, it is not serious for the majority of functional groups. In the course of this study, the opportunity has also been taken to report OPLS-AA nonbonded parameters for primary alkyl halides, halobenzenes, and many ions.

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